

Minor Project Guidelines

Exploratory Data Analysis (EDA) on Sales Dataset

Project Title

Exploratory Data Analysis (EDA) on Diwali Sales Dataset

Objective

The objective of this minor project is to help students build a strong foundation in data analysis, which is an essential step in Artificial Intelligence and Machine Learning (AIML).

This project focuses on:

- Understanding real-world datasets
 - Cleaning and preprocessing raw data
 - Identifying patterns, trends, and anomalies
 - Generating insights using data visualizations
 - Preparing data for future Machine Learning tasks
-

Tools and Libraries

- Python (Jupyter Notebook or Google Colab)
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
-

Step-by-Step Instructions

1. Dataset Collection

- Download a sales or customer-related dataset
- The dataset can be sourced from the platforms listed below

- Students may choose any dataset suitable for Exploratory Data Analysis
-

2. Data Import and Export

- Import the dataset using Pandas
 - Perform data cleaning and preprocessing
 - Export the cleaned dataset as `diwali_cleaned.csv`
-

3. Data Cleaning

- Remove unnecessary columns
 - Rename columns for clarity
 - Handle missing or null values
 - Remove duplicate records
-

4. Data Preprocessing

- Convert data types where required (for example, purchase amount to numeric)
 - Identify and handle outliers
 - Encode categorical variables if needed
 - Ensure the dataset is ready for analysis
-

5. Exploratory Data Analysis (EDA)

Create at least 10 meaningful visualizations, including but not limited to:

- Gender-wise purchase analysis
 - Age group versus purchase amount
 - State-wise sales analysis
 - Occupation versus purchase behavior
 - Product category popularity
 - Marital status versus purchase trends
 - Top 10 cities by total sales
 - Purchase amount distribution
 - Boxplot comparison by gender
 - Any additional relevant visualization based on data understanding
-

Project Deliverables

1. Cleaned dataset file: `diwali_cleaned.csv`
 2. Jupyter Notebook (`.ipynb`) or Google Colab link
 3. One-page project report (PDF or DOCX), including:
 - Project title and objective
 - Tools and libraries used
 - Screenshots of key visualizations
 - Summary of insights
 - Challenges faced and solutions applied
-

Datasets for Project

Students may use datasets from the following sources:

1. <https://archive.ics.uci.edu/datasets>
 2. <https://www.kaggle.com/datasets>
-

Web Resources

Students can refer to the following learning resources for additional understanding:

1. <https://aman.ai>
 2. <https://course.elementsofai.com/>
 3. <https://www.deeplearning.ai>
 4. <https://developers.google.com/machine-learning/crash-course/>
 5. <https://ml-course.github.io/master/intro.html>
-

Dataset Flexibility

The same project structure and EDA approach can be applied to other datasets within the same domain, such as retail sales, e-commerce customer data, or seasonal sales datasets. Only the dataset will change; the overall workflow and methodology will remain the same.

Submission Deadline: 14th January, 2026

Assigned by Ayush Kumar Srivastava (Mentor)